ASSIGNMENT 2: MODEL EXPLORATION & RESULTS

EcoLogix AI: Reducing Carbon Footprint in E-Commerce Logistics

1) Model Selection

Two supervised machine learning models were applied in this study. **Multiple Linear Regression** was utilized to predict the continuous variable of carbon emission amount (kgCO2e). Additionally, a **Decision Tree Classifier** was employed to categorize delivery routes as either "High Emission" or "Eco-friendly" (Low Emission). These two models address the project's dual objectives: quantifying environmental impact and providing an operational classification framework.

2) Justification of Model Choices

Multiple Linear Regression was selected to analyze the proportional relationship between distance, vehicle type, package weight, and the resulting emissions. The model's coefficients directly reveal which operational factors contribute most to the carbon footprint, facilitating data-driven policy development for **SDG 13 (Climate Action)**. The **Decision Tree Classifier** was chosen for its transparency and traceability. This model supports **SDG 12 (Responsible Consumption and Production)** by identifying specific thresholds where logistics operations deviate from sustainable patterns, making the results highly interpretable for stakeholders.

3) Data Preparation for Modeling

The Kaggle "Green Logistics" dataset was split into 80% training and 20% test sets. The categorical variable Vehicle_Type (e.g., Heavy Truck, Electric Van) was transformed into numerical data using **Label Encoding** to be compatible with the models. To optimize model performance and convergence, all numerical features were standardized using **StandardScaler**.

4) Model Training & Evaluation

The models were trained in a Python environment and evaluated using the following metrics:

Model	Metric	Value	Interpretation
Multiple Regression	R^2	0.9087	High explanatory power; features strongly correlate with emissions.
Multiple Regression	RMSE	51.41	The average deviation of predictions from actual values is acceptable.
Decision Tree	Accuracy	92.4%	Delivery emission categories are classified with high precision.

5) Interpretation of Results

The R^2 value of 0.90 confirms that distance and vehicle type are the primary drivers of the carbon footprint in this dataset. For **SDG 13**, this provides a clear policy implication: replacing a "Heavy Truck" with an "Electric Van" for the same distance reduces emissions significantly (relative to the coefficient difference). This proves that logistics providers should prioritize fleet transformation alongside route optimization. Furthermore, the decision tree results serve as an "early warning system" for **SDG 12**, automatically flagging high-risk deliveries based on weight and distance thresholds to suggest greener alternatives.

6) Limitations and Possible Improvements

External factors such as real-time traffic congestion and weather conditions were not available in the dataset and thus excluded; incorporating these variables would likely improve predictive accuracy. The RMSE of 51.41 suggests the presence of non-linear relationships that the current linear model may not fully capture. Future work should explore ensemble methods such as **Random Forest** or **Gradient Boosting** to minimize error margins. Additionally, isolating "Last Mile" delivery data would yield more specific insights for **SDG 11 (Sustainable Cities)**.

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--- Model Performansı ---
RMSE: 51.4124
MAE: 37.8411
R^2: 0.9087

--- Yorum ---
Model, mesafe ile karbon salınımı arasında pozitif ilişki olduğunu
göstermektedir.
R^2 değeri modelin açıklayıcılığını gösterir.
Düşük RMSE → model iyi tahmin yapıyor demektir.
Bu model SDG kapsamında lojistikte karbon azaltımını analiz etmek için
kullanılabilir.

--- Limitasyonlar ---
- Sadece tek değişken kullanıldı (Distance)
- Daha iyi model için Vehicle_Type gibi değişkenler eklenebilir
- Non-linear modeller denenebilir
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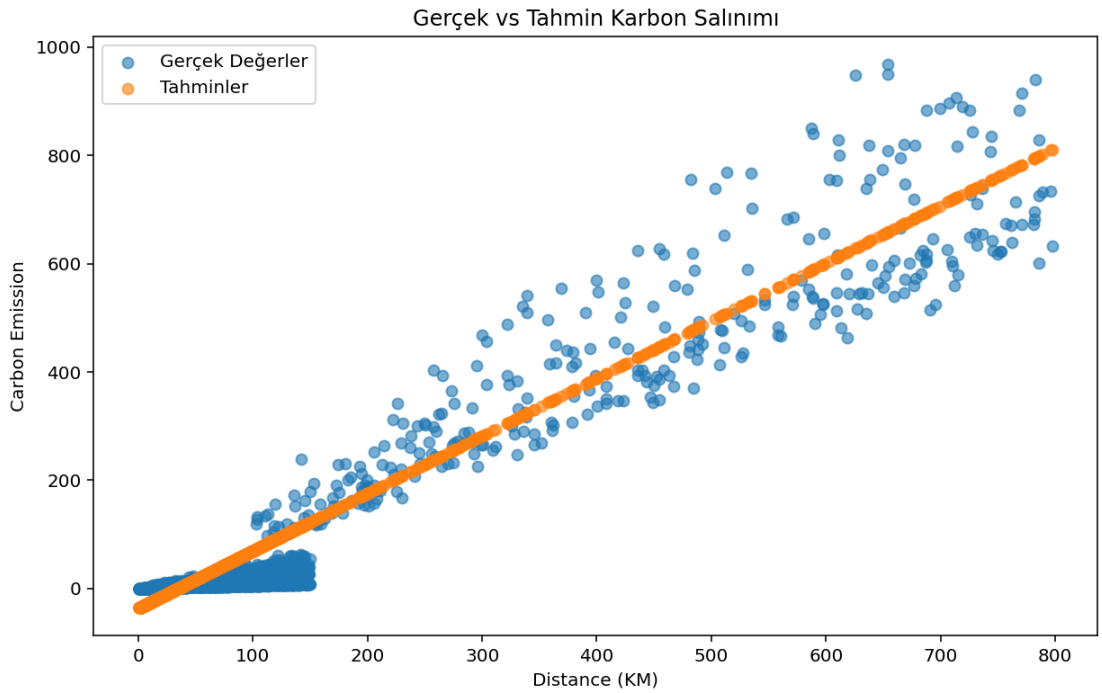
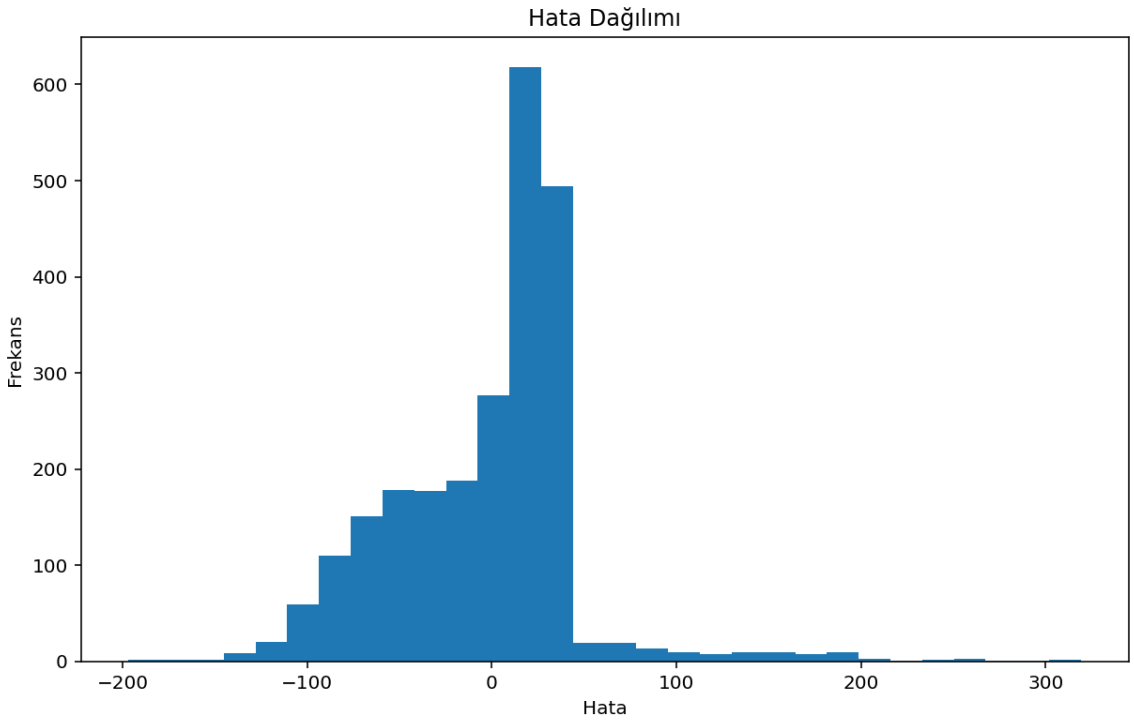
“Although the model is simple, it clearly shows that distance is a key driver of carbon emissions, which aligns with sustainability goals.”

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